

## Solution Tut - 7

Ques 1 Set  $A = \{1, 2, 3, 4\}$

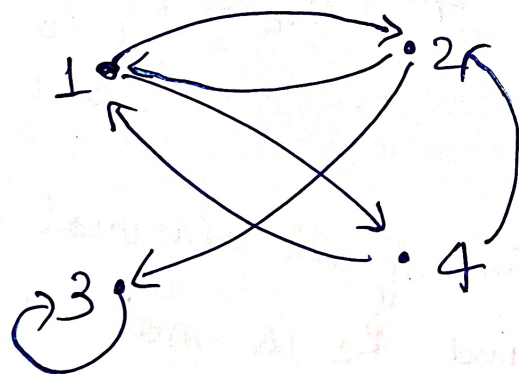
define the relation matrix  $M_R = [m_{ij}]$

where  $m_{ij} = \begin{cases} 1, & \text{if } (i, j) \in R \\ 0, & \text{otherwise} \end{cases}$

Hence the relation matrix is :

$$M_R = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

Digraph :

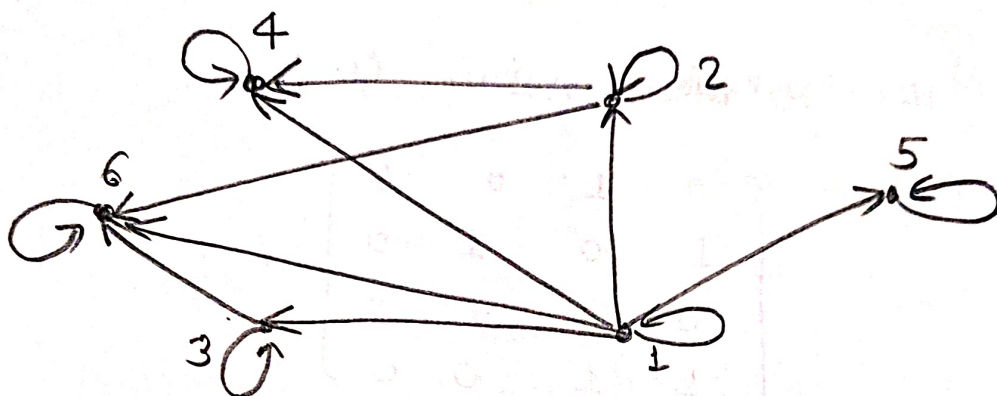


Que 2

$$A = \{1, 2, 3, 4, 5, 6\}$$

$$R = \{(a, b) ; a \text{ divides } b\}$$

$$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 2), (2, 4), (2, 6), (3, 3), (3, 6), (4, 4), (5, 5), (6, 6)\}$$



Que 3

$$M_{R_1} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \quad \& \quad M_{R_2} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

(a)

Properties of  $R_1$  &  $R_2$

① Reflexive:

A relation is reflexive if all diagonal entries are 1.  
 $\Rightarrow R_1$  is reflexive and  $R_2$  is not reflexive.

② Symmetric: A relation is symmetric if  $m_{ij} = m_{ji} \forall i, j$

In  $M_{R_1}$ ;  $m_{12} = 1$  but  $m_{21} = 0$   
 $\Rightarrow R_1$  is not symmetric.

In  $M_{R_2}$ :  $m_{12} = m_{21} = 1$   
 $m_{23} = m_{32} = 1$

Hence,  $R_2$  is symmetric.

### 3) Antisymmetric:

A relation is antisymmetric if  
 $(i, j) \in R$  and  $(j, i) \in R \Rightarrow i = j$

In  $R_1$ , there is no pair  $(i, j)$  with  $i \neq j$  such that  
both  $m_{ij} = 1$  &  $m_{ji} = 1$  ~~or~~

therefore  $R_1$  is antisymmetric.

In  $R_2$ ;  $m_{12} = 1$  and  $m_{21} = 1$  but  $1 \neq 2$   
 $\Rightarrow R_2$  is not antisymmetric.

### 4) Transitive:

$\rightarrow R_1$  is transitive as  
 $M_{R_1} + M_{R_1}^2 + M_{R_1}^3 = M_{R_1}$

$\rightarrow R_2$  is not transitive as  
 $(1, 2)$  &  $(2, 1) \in R_2$  but  $(1, 1) \notin R_2$

Ques 3 (b)  $R_1^C$

Complement is obtained by replacing 1 with 0 and 0 with 1

$$M_{R_1}^C = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$M_{R_1 \cup R_2} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

$$M_{R_1 \cap R_2} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

$$M_{R_1 \circ R_2} = M_{R_2} \cdot M_{R_1} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \\ = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Ques 4  $S = \{1, 2, 3, 4, 5\}$

$$R = \{(1, 2), (2, 2), (2, 3), (5, 4)\}$$

reflexive closure of  $R = R^* = R \cup I_S$

where  $I_S = \{(a, a), a \in S\}$

$$\Rightarrow I_S = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5)\}$$

$$\therefore R^* = \{(1, 1), (1, 2), (2, 2), (2, 3), (3, 3), (4, 4), (5, 4), (5, 5)\}$$

Symmetric closure:

$$R^{-1} = \{(2,1), (2,2), (3,2), (4,5)\}$$

$$\text{Symmetric closure} = R^S = R \cup R^{-1}$$

$$\therefore R^S = \{(1,2), (2,1), (2,2), (2,3), (3,2), (4,5), (5,4)\}$$

Transitive closure:

$$R = \{(1,2), (2,2), (2,3), (5,4)\}$$

Transitive closure is the smallest transitive relation containing  $R$ .

$$R^2 = R \circ R$$

$$\rightarrow (1,2) \text{ and } (2,2) \rightarrow (1,2) \text{ (already present)}$$

$$\rightarrow (1,2) \text{ and } (2,3) \rightarrow (1,3) \text{ (add)}$$

$$\rightarrow (2,2) \text{ and } (2,3) \rightarrow (2,3) \text{ (already present)}$$

No further chain from  $(5,4)$

$$\text{Hence } R^2 = \{(1,2), (1,3), (2,2), (2,3)\}$$

Similarly

$$R^3 = R \circ R^2 = \{(1,2), (1,3), (2,2), (2,3)\}$$

$$R^4 = R^5 = \{(1,2), (1,3), (2,2), (2,3)\}$$

$$R^T = R \cup R^2 \cup R^3 \cup R^4 \cup R^5$$

$$= \{(1,2), (1,3), (2,2), (2,3), (5,4)\}$$

Ques 5

$$A = \{1, 2, 3, 4\}$$

$$R = \{(1,1), (1,3), (2,2), (2,4), (3,1), (4,2)\}$$

$$M_R = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$